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INVITED REVIEW

Epidemiology, survival, costs, and quality of life in adults with short bowel syndrome

Marion Winkler PhD, RD[1] | **Kelly Tappenden PhD, RD**[2]

[1]Department of Surgery/Nutrition Support, Rhode Island Hospital, Warren Alpert Medical School of Brown University, Providence, Rhode Island, USA

[2]Department of Kinesiology and Nutrition, University of Illinois at Chicago, Chicago, Illinois, USA

Correspondence
Marion Winkler, PhD, RD, Rhode Island Hospital, Warren Alpert Medical School of Brown University, 593 Eddy St, NAB, 218, Providence, RI 02903, USA.
Email: mwinkler@lifespan.org

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Abstract

Short bowel syndrome (SBS) is a rare disorder with known physical, psychosocial, and economic burdens and significant morbidity and mortality. Many individuals with SBS require long-term home parenteral nutrition (HPN). The incidence and prevalence of SBS is difficult to determine because it is often based on HPN usage and may not account for those who receive intravenous fluids or achieve enteral autonomy. The most common etiologies associated with SBS are Crohn's disease and mesenteric ischemia. Intestinal anatomy and remnant bowel length are prognostic for HPN dependency, and enteral autonomy confers a survival advantage. Health economic data confirm that PN-related costs are higher for hospitalizations than at home; yet significant healthcare resource utilization is necessary for successful HPN, and patients and families report substantial financial distress that impacts quality of life (QOL). An important advancement in QOL measurement is the validation of HPN- and SBS-specific QOL questionnaires. In addition to the known factors negatively impacting QOL, such as diarrhea, pain, nocturia, fatigue, depression, and narcotic dependency, research has shown that the volume and number of PN infusions per week is associated with QOL. Although traditional QOL measurements describe how underlying disease and therapy influence life, they do not assess how symptoms and functional limitations affect the QOL of patients and caregivers. Patient-centered measures and conversation focused on psychosocial issues helps patients with SBS and HPN dependency better cope with their disease and treatment. This article presents a brief overview of SBS, including epidemiology, survival, costs, and QOL.

KEYWORDS
health care costs, home parenteral nutrition, quality of life, resource allocation, short bowel syndrome, survival

BACKGROUND/INTRODUCTION

Short bowel syndrome (SBS) is a disorder resulting from extensive surgical resection of the small intestine for conditions including Crohn's disease (CD), mesenteric ischemia (MI), intestinal trauma, neoplasm, bowel obstruction, inflammation from mucosal diseases, or radiation injury. The syndrome is characterized by severe malabsorption associated with abdominal pain diarrhea, dehydration, and malnutrition. A substantial number of patients with SBS require intravenous fluids (IVFs) or parenteral nutrition (PN). Depending on the volume and length of IVF/PN support and the degree of intestinal adaptation, they can be classified as having acute or chronic intestinal failure (CIF) or intestinal insufficiency.[1] Dependency on home PN (HPN) results not only in physical burdens associated with SBS, but also emotional, technology-, and financial-related burdens that impact quality of life (QOL). Coping strategies, including education, self-management, and peer support, are vital to help patients and their families adapt to the altered lifestyle associated with SBS and HPN so they can live and thrive. This article presents an overview of adult SBS including epidemiology, effects on survival, costs, and QOL and represents an update from our previous review.[2]

EPIDEMIOLOGY

Identifying the true prevalence of SBS is difficult because of variations in clinical classifications, variations in healthcare database information, and geographical variation in healthcare provision. As such, the SBS population has been approximated based on the prevalence of HPN use. Despite these limitations, it can be confidently concluded that HPN use has remained rare since the advent of this outpatient therapy 50 years ago.[3] Howard and associates estimated that 40,000 Americans received HPN in 1992.[4] In 2013, Mundi and associates estimated that 25,011 individuals received HPN.[5] Similarly, a recent report identifies 24,048 individuals receiving HPN between 2012 and 2020.[6] Among this most recent report, more individuals with SBS are women, are >45 years of age, receive care from providers with few patients like themselves, and travel a large distance for care if they live in rural areas.[6] However, when considering these data, it must be recognized that the use of PN use estimates those with intestinal failure, including those living with motility disorders, mucosal diseases, and other diagnoses, and therefore overestimates the SBS population. Conversely, some patients with SBS may die following resection; manage their disease with IVFs, infusion therapies, or rescue interventions; or have achieved enteral autonomy and therefore are not included in estimates of PN users. The specific incidence of SBS in the United States (US) and other countries is reported in Table 1. However, it should be noted that these data shown are only estimates because of the difficulties identifying this population, which are further compounded by methodological differences regarding data expression (point vs period data) and the collection of data during different time periods.

TABLE 1 Geographical prevalence estimates of home parenteral nutrition

Country	Prevalence estimate per million
Argentina[7]	20
Denmark[8]	19
Finland[9]	9
Germany[10]	34
Ireland[11]	10–15
Spain[12]	6
The Netherlands[13]	12
United Kingdom[14]	21
United States[6]	75

With few exceptions,[15] the top two reported etiologies leading to SBS are MI and CD. MI and CD accounted for 58% (MI 21%, CD 38%) in a United Kingdom (UK) cohort,[16] 60% (MI 35%, CD 24%) in a cohort from Denmark,[17] 49% (MI 43%, CD 6%) of a French cohort,[18] and 53% (MI 29%, CD 24%) of an Italian cohort.[19] Similarly, an international survey of 22 countries revealed that CD and MI were the underlying diagnosis in 54% of 1880 cases of SBS (MI 27%, CD 27%).[20] In this large sample, surgical complications accounted for 17%, radiation enteritis for 6%, volvulus for 4%, and a wide variety of other etiologies for the remainder.[20]

The classification of SBS based on remnant anatomy represents a predictable manner to determine severity, management, and prognosis among individual patients. The international survey indicates among individuals with SBS, 60% have a jejunostomy (SBS-J) or type I, most commonly caused by CD; 31% have jejuno-colic anastomosis (SBS-JC) or type II, most commonly caused by MI; and 9% have a jejuno-ileal anastomosis with an intact ileocecal valve and colon remaining (SBS-JIC) or type III, most commonly caused by surgical complications.[20] Though sex, body mass index, and age were similar among the three types of SBS, long-term dependence on PN is most expected with type I compared with type II anatomy. Patients with type III anatomy are typically considered as having a high probability of achieving enteral autonomy. Among all patients, the duration of HPN was <3 years in approximately half of patients but >10 years in 17%. Among those patients who were dependent on PN/IVF, the mean infusion requirements were as follows: 14.2 L/week, 6.1 days/week in type I; 11.2 L/week, 5.5 days/week in type II; and 9.8 L/week, 5.2 days/week in type III.[20]

In a cohort report from France, long-term HPN is required in 47% of patients with SBS started on this therapy.[18] Among 268 patients with nonmalignant SBS, HPN dependence probabilities were 74%, 64%, and 48% at 1, 2, and 5 years, respectively. For those patients who achieved enteral autonomy, 54.5% did so within 12 months, 19% between 12 and 24 months, and 26.5% did so after >2 years following SBS diagnosis. Factors predicting enteral autonomy were an early (<6 month) plasma citrulline concentration >20 μmol/L, >75 cm residual small bowel length, and >57% colon-in-continuity. Consistent with other reports,[20] long-term HPN dependence was greatest in patients with type I, followed by type II, and then type III residual anatomy.

SURVIVAL

In the US, healthcare utilization related to SBS remains high. Using the US Nationwide Inpatient Sample database from 2005 to 2014, Siddiqui and colleagues[21] identified adult SBS hospitalizations using *International Classification of Diseases, Clinical Modification* codes. Among 50,040 hospitalizations, SBS-related hospitalizations increased 55% between 2004 ($n = 4037$) and 2015 ($n = 6265$). During this period, in-hospital mortality decreased from 40 to 29 per 1000 hospitalizations, resulting in an overall reduction of 27%. Older patients with SBS and sepsis (6.7%), liver dysfunction (6.2%), severe malnutrition (6.0%), and metastatic cancer (5.4%) had the highest risk of mortality. During this time, the length of stay for SBS-related hospitalizations steadily decreased (the overall mean was 14.7 days), whereas the cost of care remained relatively stable ($34,130 mean hospital cost).

Not surprisingly, patients with SBS have reduced long-term survival. Initial reports in 1999 indicated that the 2- and 5-year survival probabilities for patients with nonmalignant SBS were 86% and 75%, respectively.[22] Fourteen years later, in 2013, investigators from these centers reported similar survival probabilities of 94%, 70%, and 52% at 1, 5, and 10 years, respectively.[18] Further analysis indicates that survival was more likely in patients <60 years of age, but less likely in those with mesenteric infarction, a history of cancer, or the presence of an end-ostomy. Ten-year survival was significantly higher in patients who achieved enteral autonomy compared with those who remained HPN dependent.[23] These data are supported by the report of a Canadian cohort wherein the 5-year survival of patients with nonmalignant/SBS was 81.9%.[24] In this group, patient survival was not impacted by known intestinal anatomy, patient age, or sex. However, a separate report from a Canadian cohort found higher mortality with patient age, PN dependence, number of hospitalizations, and use of immunosuppressants.[25]

Comparing patients with SBS with those in the general population has been less frequently studied. A recent analysis of UK data indicates that individuals with SBS had a 6.8-fold higher mortality rate than for the general UK population.[26] Among all forms of CIF (among which 64% had SBS) life expectancy was 17 years shorter than the general UK population.[26] Patients who did not achieve nutrition autonomy had an increased likelihood of death compared with patients who ceased HPN, further underscoring that enteral autonomy is a survival advantage.[26]

COSTS

Adults who need HPN experience substantial economic burden. Part of the financial distress relates to medical costs, nonreimbursable expenses, lost wages, fear of health insurance coverage loss, and shrinking capitation for catastrophic healthcare. Much of the stress related to financial worries likely contributes negatively to QOL. Information related to the economic burden of HPN, SBS, treatment-related expenses, and hospitalizations has been reported. Overall, being at home is more cost-effective than receiving PN in the hospital, yet the intangible costs to the individual and family are difficult to quantify.

Economic evaluations typically focus on direct costs of healthcare and nonhealthcare expenditures related to diagnosis, treatment, medical follow-up, transportation, equipment and supplies, and productivity costs in terms of losses due to morbidity and mortality. Additionally, there are intangible costs related to the social, emotional, and human costs of disease and treatment. The most comprehensive analysis of the economic costs of HPN were reported in a systematic review by Arhip et al.[27] These researchers examined existing scientific literature of partial or full economic evaluations associated with HPN. The analysis includes 21 papers that studied partial economic evaluation and two papers reporting on full economic evaluation. Of the partial economic evaluation papers, 16 were studies of cost of illness and five were cost analyses; the two full economic evaluation studies were both cost-utility analyses. Studies included in the systematic literature review were reported between 1980 and 2019, included both adult and pediatric HPN recipients, and had global representation with the majority completed in the US and the remainder from Canada, the Netherlands, Spain, France, and Belgium. Although it is difficult to draw conclusions because of the heterogeneity of studies from the standpoint of year of study, geographical location, and data included in analysis, a few important factors can be highlighted. Overall, the cost of HPN in adult patients in the US annually ranged from $103,000 to $176,000 (based on costs in 1992 and 1996), with a range of $19,000–$253,000 per patient per year (1982–1983 costs) among the included studies. The cost of HPN in Europe ranged from $13,943 per year (2017 costs converted from € to US dollars 2023) to $76,152 (2015 costs converted from € to US dollars 2023) with a notable downward trend after the first year of therapy. HPN was found to be a cost-savings compared with hospital-based PN among studies analyzing these expenditures. Similarly, the two full economic evaluations found HPN to be 65% more cost-effective than hospital care.

A nationwide analysis of the costs of adult HPN from 2010 to 2020 in Poland reported >€146 million with the total cost of health-related reimbursement for patients receiving HPN of €242 million.[28] During the decade of observation, the total costs of healthcare per patient receiving HPN decreased, yet the mean cost of health per citizen in Poland was on average 36.8 times lower than healthcare costs per patient receiving HPN. Costs of HPN and total healthcare costs in 2020 per patient were $10,838 and $17,201 (converted from € to US dollars 2023), respectively.

A retrospective single-center analysis of in-hospital costs for adult patients with type III intestinal failure in Australia found high inpatient costs that substantially outweighed current reimbursement.[29] In this report, the median length of hospitalization was 32 days (interquartile range, 20–67) with a median total cost per patient per admission of $36,675 (interquartile range, $23,196–$67,439). The main contributors to hospital costs were staff resources, time, and the cost of PN. Global implications for studies like this relate to the lack of HPN services in some countries as well as the geographical disparities in existing healthcare programs and expertise.

In three papers published in 2010, Smith and colleagues reported data from 80 families and outlined a broad array of health services used to manage HPN.[30–32] Families had between two and 20 annual physician office appointments and an average of 14 visits with allied health and complementary care (acupuncture, massage therapy, or chiropractic) professionals, between two and 26 in-home health visits, and many outpatient hospital services as well as hospitalizations for catheter-related bloodstream infections, central line replacement, disease-related problems, and fluid and electrolyte imbalances. These families attributed >10% of their annual income to cover over-the-counter medications and supplies, copays for prescriptions and physician visits, and other nonreimbursable costs. The financial distress arising from these obligations was significant, as was the need to obtain medical disability and the impact caregiving had on caregiver employment. The subsequent toll on caregivers' mental health status, depression, and daytime sleepiness cannot be calculated.

Siddiqui et al. analyzed the Nationwide Inpatient Sample database from 2005 to 2014 to evaluate trends in SBS-related hospitalizations and in-hospital mortality and to estimate the healthcare burden associated with SBS-related hospitalizations in the US using adjusted costs for inflation.[21] The report includes a study sample size of 53,040 hospitalizations by patients with SBS. Medicare was the primary insurance for 52.4% of patients whereas 31.2% were privately insured. The overall mean cost for SBS hospitalizations was $34,130 (2014 dollars), with costs of care significantly higher for patients with sepsis, liver disease, and severe malnutrition.

Mundi et al. examined the Symphony Health Solutions' Integrated DataVerse open claims database that consists of longitudinal data for pharmacy claims, physician office medical claims, and hospital claims specific to patients with CIF with at least two parenteral support prescriptions within 6 consecutive months from October 2012 to June 2020.[6] Of the 24,048 patients equivalent to an estimated prevalence of 75 patients with CIF per million US inhabitants, 56% had commercial health insurance, 33% had Medicare, and 10% had Medicaid. The analysis identified a disparity in healthcare coverage in rural vs urban areas, few providers managing large patient groups long-term, and approximately half of the patients traveling between 10 and 100 miles for HPN-related management.

Loutfy et al. analyzed a commercial database to identify prescribed teduglutide from 2015 to 2019.[33] Of the 72 million patients in the database during the study period, 170 patients were prescribed teduglutide, the majority of whom had either Medicare (41.2%) or Medicaid (41.2%). As the indication for teduglutide is SBS and parenteral support dependency, most patients had a coded diagnosis of postsurgical malabsorption. Costs associated with teduglutide treatment or healthcare were not reported. In a cost-effectiveness study of teduglutide in adult patients with SBS in the US, Raghu et al. concluded that teduglutide becomes economically reasonable only if its cost is substantially reduced.[34] They cite an estimated cost >$400,000 per patient per year compared with PN dependence of $150,000 per year.

The Oley Foundation (http://www.oley.org) and other like organizations worldwide are committed to advocacy efforts through the encouragement of self-advocacy, a call-to-action for legislative and public policy issues, support for access to equitable treatment, and outreach to government agencies for increased research funding. Efforts to resolve the psychosocial stress resulting from the paradox of being unable to work and the need for disability insurance will hopefully positively influence QOL.

QOL

Good QOL as defined by the World Health Organization is a state of complete physical, mental, and social well-being that is not shaped solely by the absence of disease (https://www.who.int/tools/whoqol). Recognizing that perceived QOL is highly subjective, an individual's perception of satisfaction and happiness in life greatly influences how it is described and rated. Historical descriptions of QOL among patients with SBS, CIF, and/or HPN dependency include a long list of negative influencing factors, such as age, low self-esteem, drug/opiate dependency, depression, poor coping skills, inability to eat, diarrhea, abdominal pain, infections, complications, polyuria, sleep disruption, and a disturbed social life.[35] Current reports recognize the resiliency of some patients and their ability to balance both positive and negative factors influencing QOL as they adapt to and cope with the consequences of SBS and the therapeutic interventions and treatment.[36] Advances in our understanding of QOL can largely be attributed to insight into patients' lived experiences and the development and validation of disease and/or therapy-specific questionnaires to measure and evaluate QOL.

A good QOL among HPN-dependent adults, the majority of whom have SBS, is associated with enjoying life; being happy, satisfied, or content with life; and being able to participate in desired daily activities whenever wanted.[37] Positive factors influencing the ability to engage in household, social, recreational, and leisure activities include feeling energetic, having physical strength and stamina, having strong support systems, and having flexible PN infusion schedules. Negative influencers include abdominal pain, diarrhea, high ostomy output or leakage, infection, disease- and therapy-related complications, recurrent hospitalizations, depression, and financial distress.[37]

An important advancement in the measurement of QOL among patients with SBS and/or HPN dependency is the conclusion that generic health status measures of QOL do not capture the clinically relevant factors of living with SBS and/or HPN dependency. This recognition led to the development and validation of HPN therapy–specific and SBS disease–related questionnaires that can be used in research and clinical practice (Table 2).

The HPN-QOL questionnaire includes 48 questions, eight functional scales, nine symptom scales, several HPN-related questions, and three global QOL questions.[38] Scored on a scale of 0–100 (poor to high QOL), the functional scales address general health, ability to travel, coping, physical function, ability to eat and drink, employment, sexual function, and emotional function. The symptom scales address body image, immobility, fatigue, sleep patterns, gastrointestinal complaints, pain, stoma presence, financial issues, and weight. The HPN-specific questions address the infusion pump and nutrition team support. Two specific QOL questions rate the degree to which the underlying illness and the degree to which HPN affect QOL. The global QOL score is rated as poor, fair, good, very good, or excellent. The questionnaire is available in English and has been translated into Danish, Dutch, French, German, Italian, Polish, and Spanish. A large international study using the HPN-QOL among 691 HPN-dependent adults in 14 countries reported better QOL if PN dependency was >24 months or if SBS was caused by CD or MI vs cancer, motility disorders, or radiation enteritis; and it reported worse QOL if the patient was living alone or had a greater number of PN infusions per week.[43] The HPN-QOL questionnaire has been modified by researchers in Israel,[40] who created a shorter version validated for use in clinical practice, and by Pironi and associates to evaluate QOL before and after intestinal transplantation.[44] When compared with other QOL instruments, HPN-dependent patients in a small cohort in Denmark found the HPN-QOL the most relatable.[45]

A disease-specific questionnaire (SBS-QoL) was developed and validated against the HPN-QOL questionnaire and used to quantify treatment-induced changes in clinical research studies using teduglutide.[39] This questionnaire includes 17 items, two subscales, pictorial illustrations, and a visual analogue scale with a 7-day recall response period. The SBS-QoL is scored on a scale of 0–170 reflecting perfect to worse QOL. Parenteral volume reduction had a beneficial but nonsignificant influence on QOL in patients receiving teduglutide compared with a placebo.[46] Subgroup and post hoc analysis was conducted to determine the impact of teduglutide on QOL by examining parenteral support volume requirements, disease etiology, and bowel anatomy. Results showed the most pronounced QOL changes among research participants who had the highest baseline parenteral support volumes and

TABLE 2 HPN- and SBS-specific QOL instruments

QOL instrument	Description/scales	Score	Reference
HPN-QOL	48 questions 8 functional scales 9 symptom scales 2 nutrition support team items 3 global quality of life scales	0–100 (poor to high QOL)	Baxter et al.[38]
SBS-QoL	17 items 2 subscales Used to evaluate treatment-induced changes in teduglutide research studies	0–170 (high to poor QOL)	Berghofer et al.[39]
New QOL, a modified short version of HPN-QOL	1 general health item 9 physical symptom items 11 level of independence with HPN administration items 2 technical/emotional support items 3 questions on effect of HPN on mental state, physical abilities, and social life 9 questions on hospitalization and use of healthcare services	Composite score not reported	Theilla et al.[40]
Parenteral Nutrition Impact Questionnaire (PNIQ)	20 items True/false Measures the effect of HPN on daily life	0–20 (good to poor QOL)	Wilburn et al.[41]
Home Parenteral Nutrition Patient-Reported Outcome Questionnaire (HPN-PROQ)	20 patient-reported outcome statements 3 – HPN therapy goals 9 – training and preparedness for HPN care 5 – health limits of disease and treatment 3 – emotional coping 3 – global quality of life	Self-assessment tool; does not compute a compositive score	Winkler et al.[42]

Abbreviations: HPN, home parenteral nutrition; QOL, quality of life; SBS, short bowel syndrome.

inflammatory bowel disease. Teduglutide was generally associated with a greater improvement in QOL, but the differences were not significant.[47] In a single-center study of patients with nonmalignant SBS and intestinal failure, parenteral volume was associated with a worse QOL score and significantly poorer QOL scores among patients with a residual small intestinal length ≤50 cm compared with those with a greater remnant small bowel length.[48]

Decreased PN infusions have also been shown to result in improved QOL scores when measured by the PN Impact Questionnaire.[49] The PN Impact Questionnaire is a 20-item true/false questionnaire that focuses on how HPN limits QOL. Validated among 30 patients receiving HPN in the UK, this questionnaire found a moderately high correlation between QOL and social isolation, emotional reaction, and energy level.[49] Scored on a scale from 0 (good QOL) to 20 (poor QOL), Burden et al. demonstrated better scores if patients infused 1–4 nights of PN per week compared with 5–7 nights per week.[49]

Having one or more nights without PN and having a shorter PN infusion schedule was very or extremely important to 66% of respondents with PN dependency and intestinal failure; however, 14% were neutral and 20% rated it low in importance or not at all important.[42] Some individuals were unaware they could have flexibility with their infusion schedules, underscoring the importance of having clinician-patient conversations about goals, priorities, and personal preferences of HPN therapy. When desired by a patient and the clinical situation allows, a reduction in PN frequency should be considered to improve QOL in patients receiving HPN.[50]

The use of patient-reported outcomes (PROs) is a clinically useful approach to monitor progression of disease or response to treatment as well as to improve clinician-patient communication and expand the role of shared or informed decision making.[51] HPN-dependent adults desire a holistic approach to their healthcare with emotional support provided in addition to medical care and nutrition assessment.[37,52] The validated HPN PRO Questionnaire (HPN-PROQ) is a clinically relevant approach to identify and address lifestyle concerns of patients receiving HPN during a medical office or homecare visit. The HPN-PROQ includes 20 statements that address HPN therapy goals, education and training preparedness for self-care, health limitations caused by disease and HPN therapy, emotional coping, and three global QOL questions. The PRO statements are rated on a Likert scale. The goal is not to compute a composite score but to be a self-assessment and conversation starter. This tool has been shown to be sensitive to QOL differences among patients with CIF and prolonged acute intestinal failure.[42] Confidence in the ability to perform HPN procedures was associated with an understanding of the need for therapy, and as ratings increased for emotional difficulty in coping with HPN so did HPN impact on QOL.[42] The HPN-PROQ helps patients prioritize important lifestyle issues they wish to discuss with their clinician, allows the individual to share how they are feeling and obtain immediate feedback, and ultimately improves the clinician-patient relationship by creating a positive patient experience.[53] Other patient-centric measures are also recommended to address information that is of primary concern and value to patients. Heaney et al. point out that although traditional health-related QOL assessments describe the potential impact of disease or therapy on a patient, they do not assess how the individual's life is influenced by symptoms and functional limitations.[54] Such information may best be elicited through interview and conversation.

The important role of having family support is well established in the QOL and HPN literature.[55] Living with SBS and HPN dependency impacts not only the patient but family members as well. Among family caregivers, sleep deprivation is common and leads to further suffering from daily distress and fatigue. Furthermore, family caregivers may neglect their own physical and mental health; have a high rate of reactive or situational depression, anxiety, and fear; and often express feelings of social isolation.[35,56] In a large multicenter study in the UK of caregivers representing 339 patients with CIF, the family members' own health status and lack of health service support was more likely to be associated with moderate to severe caregiver burden.[57] Male sex is associated with a higher caregiver burden,[58] and family caregivers of patients with intestinal dysmotility compared with SBS experience more caregiver strain because of demands on their time.[59] Resources for patients and their family caregivers should be provided to facilitate coping with SBS and HPN dependency. Economic stress or financial distress related to nonreimbursed costs of healthcare, inadequate insurance coverage, and lost wages caused by an inability to work also contributes to poor QOL among HPN-dependent individuals and to their caregivers' mental health status and income adequacy, depression, and fatigue.[32]

Now that there are a number of validated therapy-specific, disease-specific, and PRO questionnaires, the need for prospective, longitudinal repeated measures studies to evaluate HPN-dependent adults with SBS and CIF and their family caregivers are important to close research gaps. Coping with the complexities of HPN and SBS/CIF requires a comprehensive and interdisciplinary approach for teaching patients and their caregivers to manage their therapy at home. This process should start immediately in the hospital prior to discharge and should continue throughout the patient's SBS and HPN trajectory. Informed and knowledgeable patients will be better able to accept the understanding of their diagnosis and the need for HPN therapy, and they will be able to gain confidence in mastering the skills necessary to help them cope and thrive at home.[60] Similarly, continuing to have close access to support will foster the security needed to deal with adjusting to a significant change at home.

It is well established that patients receiving HPN who participate in organizations that provide ongoing education and peer support have significantly better QOL scores compared with nonaffiliated patients.[61] Similarly, Chopy et al.[62] found that competency, inspiration, normalcy, and advocacy were key themes in how patients receiving HPN and home enteral nutrition described the value of membership in The Oley Foundation. Future research should explore whether in-person or online support group attendance improves QOL. To date, there has been limited research of the HPN and CIF population that directly studies this relationship; however, an innovative family-centered strategy using video conferencing provided patients receiving HPN and caregivers the opportunity to gain peer support in real time[63] and introduced participants to online resources and preloaded applications to address mood, hope, breathing, power naps, mindfulness, resilience, self-monitoring, goal setting, and problem solving. Satisfaction was very highly rated, and some participants expressed the sentiment that these sessions were life-changing.[63] Similarly, a web-based mindfulness program for patients with benign CIF who receive HPN was subjectively associated with improved QOL.[64]

CONCLUSION

SBS is a complex, debilitating disease that affects a relatively small population; however, the burden is great by compromising patient QOL, increasing morbidity and mortality, and significantly using healthcare resources. Available data indicate that many of these factors have not changed over the past 25 years. Improved outcomes may await care models wherein more patients with SBS receive specialized interdisciplinary care from established intestinal rehabilitation programs with multidisciplinary teams of providers with the goal of fostering intestinal adaptation such that dependence on PN can be decreased.[65]

AUTHOR CONTRIBUTIONS

Marion Winkler and Kelly Tappenden equally contributed to the conception and design of the research; Marion Winkler and Kelly Tappenden contributed to the design of the research; Marion Winkler and Kelly Tappenden contributed to the acquisition and analysis of the data; Marion Winkler and Kelly Tappenden contributed to the interpretation of the data; and Marion Winkler and Kelly Tappenden drafted the article. Both authors critically revised the article, agree to be fully accountable for ensuring the integrity and accuracy of the work, and read and approved the final article.

CONFLICT OF INTEREST STATEMENT

Marion Winkler serves as an Independent Nutrition Consultant for VectivBio. Kelly Tappenden serves on the clinical advisory board and as speaker for Takeda and VectioBio and as a speaker for Nutricia and Abbott.

ORCID

Marion Winkler http://orcid.org/0000-0003-0946-8294

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